Transparent AI Framework for Identifying and Explaining Claim Denials in Healthcare Using Independent Medical Review (IMR) Data

Capstone Project Proposal

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Abstract

This proposal presents an Explainable Artificial Intelligence framework for predicting and explaining healthcare claim denial outcomes using the California Independent Medical Review dataset. The study combines healthcare analytics, machine learning, and SHAP-based explainability.

Introduction

Healthcare claim denials create operational challenges, reimbursement delays, and administrative costs. This project investigates whether transparent machine learning models can predict review outcomes while providing understandable explanations.

Background and Motivation

Healthcare organizations process thousands of claims that may be denied, delayed, or modified. Explainable AI can improve trust and transparency in decision support systems.

Problem Statement

Most predictive models provide outcome predictions but do not explain why a claim is likely to be upheld or overturned.

Objectives

Analyze IMR data; build prediction models; compare algorithms; implement SHAP explainability; evaluate transparency and performance.

Literature Review

Prior research demonstrates the effectiveness of Random Forest, XGBoost, and explainability methods such as SHAP in healthcare analytics. Responsible AI frameworks emphasize transparency and accountability.

Dataset Description

The project uses the California Department of Managed Health Care Independent Medical Review Determinations dataset containing historical IMR decisions.

Methodology

Data collection, preprocessing, exploratory data analysis, feature engineering, machine learning, explainability analysis, evaluation, and reporting.

System Architecture Diagram

IMR Dataset -> Data Cleaning -> Feature Engineering -> ML Models -> Prediction Engine -> SHAP Explainability -> Human Readable Explanations

Workflow

Problem Definition -> Data Collection -> Preprocessing -> EDA -> Modeling -> Explainability -> Evaluation -> Reporting
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Machine Learning Models

Logistic Regression, Decision Tree, Random Forest, and XGBoost will be evaluated using Accuracy, Precision, Recall, F1 Score, and ROC-AUC.

Explainable AI Framework

SHAP will be used for global and local explanations, feature importance analysis, and transparency reporting.

Sprint Plan

Sprint 1: Data collection and EDA.

Sprint 2: Preprocessing and feature engineering.

Sprint 3: Model development.

Sprint 4: Explainability.

Final stage: evaluation and reporting.

Gantt Chart

Oct: Proposal and EDA. Nov: Data preparation. Dec: Modeling and SHAP implementation. Jan: Final report and presentation.

Expected Outcomes

A transparent healthcare AI solution that predicts IMR outcomes and explains the reasoning behind predictions.

Ethical Considerations

The project will focus on transparency, fairness, accountability, privacy, and responsible use of healthcare data.

Limitations

Public datasets may contain limited features and may not represent all healthcare scenarios.

References

California DMHC IMR Dataset; SHAP literature; Interpretable Machine Learning; Responsible AI guidelines.